

ActInf Livestream #043.2 ~ "Predictive Coding: a Theoretical and Experimental Review"

Livestream | 1:57:24

Hello everyone.

Welcome to ActInfLab livestream number 43.2.

It is May 11th, 2022.

Welcome to the ActInfLab.

We are a participatory online lab that is communicating, learning, and practicing applied active inference.

You can find us at the links on this slide.

This is a recorded and archived livestream, so please provide us with feedback so we can improve our work.

All backgrounds and perspectives are welcome and will follow good video etiquette for livestreams.

If you want to learn more about what's happening at ActInfLab, greetings.

Head over to activeinference.org and you can find out more about participating on livestreams or other modes of participation.

And today we're in 43.2 where, as always, it's up to the last minute with what we actually do.

And welcome also to Stephen.

We're going to continue this discussion in our third part of the conversations we've been having on this paper, Predictive Coding, A Theoretical and Experimental Review.

And we'll begin with the introductions.

And then we can move around the paper.

We brought up some things that we'd like to cover last week for today.

But also we can take any questions or areas or sections that any of the stream want to talk more about.

We can absolutely go there.

So we'll just start with introductions and say hi and feel free to add anything you'd like.

What brought you to the paper or what's just one thing that you took away from it?

And then I'm sure we'll have some fun today.

So I'm Daniel and I'm a researcher in California and I'll pass it to Dean.

How goes it, Dean?

Oh, sorry, I was muted in Jitsi.

So it was on the live stream.

But welcome to Dean and Stephen.

And I'll also share the slides with you here in the chat.

But we're live and in the game.

So feel free to say hello and anything just to get us going with this discussion.

I'm getting a bunch of feedback from somewhere, Daniel.

It's going good.

I'll open up another window so that you can see what slides I'm working on.

But yeah, how goes it or what was interesting to you about this paper?

Do you hear me, Dean?

That's me.

Okay.

All right.

okay all right all right okay i can do you know i can hear you now yeah i could hear you i couldn't hear you before something was weird sound fine dean i can hear you daniel but there's a long delay and uh there was a i can hear i could hear feedback like a second audio loop going off while you were talking okay did you have the stream open or anything i just no i just have the picture of the of the three of us okay i'll just reload just to be sure so it goes with live streams all right okay so is that better we'll

roll with it then so okay or we'll see what happens so yeah stephen how goes it and what did you like about the paper as we start

well i'm just interested in the general scope um as per normal so i'm i'm sort of going to go with the go with the the overall vibe of um pushing the boundaries of active so um i'm fairly i'm fairly relaxed with this one okay hey dean work better yeah my my bad all good all good thank you for telling me not to have both jitsi and the youtube open at the same time so yes uh that uh these little technical things that i should have been uh more on top of there's like some reflex where if your own audio is played back to you at a certain delay it makes it like basically implausible to speak it's pretty interesting yeah well apologies guys that's not the start i wanted all good so how goes it or what like brought to you to this paper or what did you find interesting and what they wrote well i'm i'm not really totally up on on what predictive coding is i think i know what predictive processing is and when when you uh and maria were talking in the dot zero or you were blue or talking in the dot one um just the sort of the translation of that into operations that you can you can um identify with a certain amount of confidence is the part that i'm really interested in i'm not really sure what the applications are beyond the psychology realm i'm sure there are quite a few and that's

kind of why i showed up today was to sort of piggyback on what what people who obviously know a lot more about this than i do are saying about it and where it's going and what the potentials of it are all right well let's start with a few reminders and recalls is there any area or figure or formalism that either of you would like to suggest that we jump to to remind ourselves of anything critical about the paper otherwise i can bring this to some is it okay to start with the back propagation of error yeah all right so i'll bring in this was in um section four so i'll be bringing in some but let's um we're gonna ramp into it just as a reminder about the roadmap so the first section of the paper is an introduction the second section of the paper describes with quite a lot of detail the core kernel of the predictive coding architecture this is a computational architecture and it involves information transmission that is in this predictive coding way so check out the paper the dot zero the dot one that's where we're going into a lot of detail on predictive coding as an information architecture and then as maria was bringing out for us predictive processing might be used to refer perhaps more holistically to systems engaging in predictive coding and there's a bi-directionality there to predictive processing whereas predictive coding might be something that could be done in a bit more of a uni-directional way that was section two on predictive coding introducing the kernel and then going into a few specific generalizations of that kernel and modifications that help make it more computationally powerful as well as linking it closer to certain biological architectures of tissues like in the retina and in the cortex of the mammal brain in predictive coding in the brain three there's some exploration on different paradigms of predictive coding and these are referring to kind of like settings where the core and elaborations can be applied and that's in the supervised and unsupervised cases as well as thinking about spatial and temporal predictive coding algorithms and connections with modern machine learning like with deep predictive coding then after that sort of kernel and generalization of the predictive coding kernel and exploring some of the paradigms for where that kernel and its elaborations have been applied in section four there's the relationships between and among algorithms and so that's where we get into this section on predictive coding and the back propagation of error so that's where we'll go and um i'll copy in some formalisms or find out um which ones would be the best to bring in but um here we are on slide 56 which i'll share in the chat here what is interesting to you about back propagation of error then we'll start there well what i'm curious about is is the that bottom line of that slide this recursion is identical to that of the propagator to the propagation of error algorithm which are not which i'd never actually encountered before this so i was wondering i i know a bit about recursion i think we all have a sort of a basic understanding of what's involved in in recursive um loops but i was wondering how sort of the predictive where's the where is this relationship between predictive coding and this other algorithm which i've heard that i've heard that that i don't know if it's an expression or label back propagation of error but i've never really looked at it from a from a operational standpoint or you know how it's how it's laid out mathematically and i'm not sure if either of you have that familiarity but

i was hoping maybe you did great so how does predictive coding relate to the back propagation of error let's um look at these sections which we didn't annotate to heavily leading up to because you know there's like many equations in the paper many areas um even before we introduced the formalism and to give me 15 seconds to read it qualitatively what would back propagation of error mean to you or where is an applied setting where you think this could matter well and again this is this is me talking out loud hanging my butt out there because i don't really know what i'm talking about but am i am i correct in assuming that it's um anytime you find yourself in feedback and feed forward loops i know that they're a cyclical and i know that usually implies a directional but isn't anytime you're getting sort of a a looping condition where you get an opportunity to sort of reset isn't that what we're isn't that the situation that we're examining okay we're in the arena of statistical bayesian graphs so we're talking about models here we're not necessarily talking about what an actual system is doing the computer system nor some sort of biological system we're talking about a modeling statistical architecture and it would be a secondary conversation about what systems are actually doing with back propagation so this isn't just like salt water flowing from the ocean into the river okay just to sort of forestall any preliminary connections we might want to draw but we can see when we have a bit more sense of what back propagation of errors on this graphical setting how it could apply so they're introducing this concept of a computational graph fancy g not free energy related just g for graph and it consists of two kinds of elements just like all graphs or networks do which is vertices and edges the vertices are representing intermediate computation products for example the activations at each level this multi-level predictive processing architecture so we have like nodes within the level and then there's edges that represent influence um here as differentiable functions so this is related to some of the graphical bayesian approaches we've looked at previously but also it shares a lot with some neural network representations where the edges are not just like a correlation or mutual information but the edges are more like a differentiable function okay we're going to restrict to looking at specific kind of graphs here which are computation graphs they're saying are directed acyclic graphs so that means that it's directed there's an arrow such that the differentiability of the function is going from one way to the other like we can think of that as the forward direction and then the acyclicity is just ensuring that there's like sort of one place where we could start and we could just forward propagate everything out to the edges and that wouldn't trap us into an infinite recursion by going forward and we've talked about some of the differences between like cyclic and acyclic graphs with majid when we were talking about the the dcm the and all the ambiguities of that acronym um okay steven anything and then we'll continue

just one quick question um ontologically how how useful or problematic is the word coding now we've moved a bit further down with active inference um and in a way is back propagation of error needed to be that needs to be distinguished out even in a descriptive term just so that the connotations of coding don't sort of confuse all the discourse um what are the connotations of coding well that it's basically kind of almost like a representational kind of um script almost that's been written somewhere um it's kind of a how you know like the coding that you get in a computer program or something um and i kind of

think that that does sort of speak to the nature in a way because it's this is the the nature of how for the future but um back propagation of error um kind of speaks in some ways to the nature of how the code is processed um and i may be off off topic here but i just i just kind of wonder as we're going through this and trying to uh link it all together um how much uh that ontological term um is is also possibly something that could be thought of in another way to make it easier to to to to to understand what's going on okay um we are dealing here with computational architectures so i don't think we need to run away from these questions that were brought up more in a philosophical non-computational version into the anti-computationalist perspective like we're not fighting for what is happening in the brain here we're using predictive coding to talk about information coding and transmission approaches so that's one thing and um yeah i mean we don't we can't just go around flipping different words predictive coding is what for decades has been referenced by a certain architecture here and then there's a lot of openness with how that architecture is applied but i i i'm not 100 sure like this is about predictive coding architectures yeah no that's cool i'm i'm just putting that out there just seeing how that floats a little bit i think that's a good good response like what else would what else could be in this position of coding it's about how signals are encoded and what is being transferred between different nodes is what is being transferred the raw value or in the predictive coding architecture we have um two directions that signals are being passed in this compute graph not in the brain and that would be coming from um one direction the expectation that's the predictive part and then the error residual that's the deviation from the expectation and so that is what the predictive coding architecture is it's just about having graphical systems that are passing expectations and differences or errors rather than passing like just the plain values themselves yeah i suppose there's also that question of um how much is in the dynamics of what's going on and the hyper parameters when looked at and um how much difference that makes to to this story okay so you've got you've got the the the the predictions uh the prediction error has been passed down um but coding tends to be like the numbers are there but in some ways the numbers can kind of evolve once the the system the dynamics of the systems or the dynamical sort of patterns of the system start to sort of fire up so to speak you know it takes a little bit time for the active inference model to start really purring along sometimes so that kind of coding is in a way sort of almost more alive than traditional coding which is kind of sort of baked in okay i got you we're not even in active inference yet so i i see some back later but we're discussing predictive coding back propagation of error and then action does get brought into the picture in section 4 5 but up until section 4 or 5 there's no action even in the model so we're just looking at the architecture that does computation and then understanding how action could be brought in and that's kind of the fun surprise that we uncovered in the dot zero and dot one and in the layout of the paper was when the architecture is sufficiently general and powerful incorporating another variable that is about action like policy selection becomes fluent and that's why these architectures are composable with different sensory modalities with different scopes of action with different affordances all these things because there's a level of generalizability that we

can pull back to okay so let's return to predictive coding and back propagation of error we have this graph the vertices are representing intermediate computational products so like a number or like a variable

because it's kind of like where you're storing the intermediate computation product and then the edge of differentiable functions um could be you know like $y = 2x$ is a differentiable function anything that has smoothness in the graph is differentiable and as the orange part says we're restricting this to

directed acyclic graphs where we're not going to get into an infinite regress just by going forward

now what happens when we're trying to tune this architecture so that it has a minimized loss function

loss function whether we're dealing with least squares error like a regular linear regression

or whether we're dealing with some other loss function the loss function is like what is trying to be

minimized so when the ball is rolling downhill we're minimizing like the loss or the energy function

but the energy function approach and wording is coming more from like a physics

side whereas the loss function is just a purely statistical claim like what is the function that we're

minimizing okay given the output vertex v_{out} so this is like the one that we actually want um to be reducing the loss around

okay the loss function is going to be L is a function of v_{out} back propagation can be performed upon a computation graph

so we're going to be minimizing loss at a certain vertex by back propagating upstream a lot of what led to it being the way it is

backpropagation and so that could be done on just the last tip of the sphere or it could be done

throughout the whole system backpropagation is an extremely straightforward algorithm

which simply uses the chain rule of multivariable calculus to recursively compute the derivatives of children nodes from the derivatives of their parents okay so formalism 28.

this fancy d is partial differential so the change of the loss function

with respect to the change in an intermediate variable i is what we're looking at we're looking at how

we're going on that bowl are we dropping on the loss function are we is the model getting better or worse

with respect to changes of intermediate computational products vertices and that overall loss function

is the sum of v the subset of p which is the parents so we're looking at v_i like the node that we're

focused on node of interest or something and we're going to sum over all of its parents and then for all of

its parents we're going to compute something that's basically like the change in loss function over the

change in that parent's value $b_{sub j}$ and how it changes with respect to change in the child so we're

taking something that's like this graphical network with nodes and edges and differentiability and then

figuring out which way is the overall downhill based upon all of the back propagation of the gradients

by starting with this output output gradient that's v_{out} um and if all the gradients are known so

like we're dealing with a model that we specified then the derivatives of every vertex with respect to

the loss can be recursively computed so because all the edges are differentiable the big function is also

differentiable so like $2x$ is differentiable x^2 is differentiable x^3 and x to the 11th power are all differentiable functions so you could also differentiate their concatenation through addition

that'd be like the derivative of $2x$ plus the derivative of this and also the chain rule in calculus allows us to like nest those so if we can differentiate $2x$ and if we can differentiate x squared we could differentiate $2x$ inside of parentheses squared that's basically the chain rule in calculus

um and then that is what allows this sort of big downhill computation based upon the specification of the graph

and then the back propagation of all the relevant nodes and finding out how those need to be tuned so that the final output node that's being optimized can be going downhill defined as going down on the loss function less loss is better here okay yes excellent thank you because this was one big opaque mark off blank to me when i read it i just couldn't make any sense of it but is the ability to differentiate or what they is what they're saying here is the ability to differentiate an ability to build relationship that parent child thing is it is it that is it the capacity to differentiate that allows for the relationship to be identified or is it the other way around because when when they're talking here and saying back propagation performed upon a computation graph that propagation is extremely straightforward and uses the chain rule of multivariate stuff when it's talking about that it's it's it's pointing back to something that you don't necessarily have to see but you know that there is some relationship just like there is between $2x$ and x squared am i getting is that is your explanation helping me move close to a better understanding of that because i understand derivatives but i'm not sure how the derivatives fit into this explanation great yeah i think we're um we're solenoidly exploring now we haven't even brought in predictive in this these equations here are for 28 and 29 are about back propagation on computation graphs so right with these papers it's really important to know like what has not been brought onto this diction and there isn't error this is just about the nodes and edges like the intermediate computation values of the abstractest kind and their differentiable relationships so now they're going to show how predictive coding relates to back propagation of error so they write predictive coding can also be straightforwardly extended to arbitrary computation graphs to do this we simply augment the standard computation graphs with additional error units for each vertex so let's recall what some of these graphs that we previously looked at here in figure one of the paper we have a graph where the nodes are intermediate computational products represent differentiable relationships and importantly some of these intermediate products have the semantics or the interpretation of being estimates about mean μ at that level or estimates about variance and error at that level ϵ so this is kind of like a subtype of graph like the abstract is kind it's just intermediate computational products and then predictive coding are specific kinds of computational graphs where some of the nodes represent mean estimates and some of them are error estimates in contrast and hopefully not to confuse here's from 26.2 where we talked about like integrator chains so notice that all of these nodes can be interpreted as intermediate computational products but there's no error term here so this is an example of a computational graph that doesn't have error terms this is not a predictive coding architecture not that it's inconsistent with or somehow disjoint from but this is not having ϵ s so it's not a predictive coding architecture whereas this is sort of a canonical multi-level predictive coding architecture great hey jacks good to see you in the chat okay now we're going to return back to the back prop

okay so we're adding in these additional error units ϵ_i for each level of the graph formally the augmented graph becomes now $\tilde{g}$ so previously fancy g was just edges and vertices nodes and edges and now the vertices is is being unpacked so that v can remain for the mean estimators which are ultimately the ones that like we would like to have the most accuracy on accounting also for variance estimators and so ϵ this big ϵ has been pulled out it's the set of all error neurons we then adapt the core predictive coding dynamics equations from a hierarchy of layers to arbitrary graphs the change in v the nodes with respect to time prediction is happening through a sequence of observations in the predictive coding architecture that's that now casting column and filter frame differencing a lot of these other sort of real-time algorithms for mean and variance estimation that we talked about equals the change in f free energy with respect to change in the node so now the loss function through time is going to be related to the f that had been previously u_m previously defined in in earlier equations and then that is going to be defined as basically the error this is in formalism 30 the ϵ_i so the the error variance at that last level minus a sum of c so c are the children and then there's just some more terms involving basically tuning the nodes the nodes accounting for the fact that we want to like tune the variance estimators and the compute computation nodes that represent the mean estimates and so u_m then the dynamics of the parameters on the vertices v and the edge functions θ so they kind of slightly evolve the edge notation towards u_m can then be derived as a gradient descent on f where f is the sum of prediction errors of every node in the graph so previously u_m they had specified how the the composability of the multi-level predictive coding architecture is such that the levels are can be factorized from each other which allows us to sum their free energy contributions into like a free energy total and so that is how a multi-level model can be fit because it's finding like the free energy minimum for that whole multi-level system which might entail one of the levels not being at its like perfect perfect lowest free energy state and in this section they've connected the way that back propagation in 28 and 29 without any action without any prediction is able to make a big loss function that's composed because of the tractability of the compute graph and all of these little loss functions can be summed into a big loss function that you use to like train a neural network for example u_m and then moving that into the predictive coding in free energy minimizing space by saying well this is a special kind of compute graph that has the mean estimators μ and the variance estimators the error terms and then we're going to do something similar where we're going to compute like what amounts to a loss function f the sum of prediction errors of every node in the graph based upon internal prediction errors being minimized so in the non-predictive coding u_m frame we're minimizing a big loss function that's composed of smaller loss functions in the predictive coding frame we're minimizing our divergence from expectations composed of the divergences from many smaller expectations okay yeah go for it so extrapolating hopefully again from this the the predictive coding then like by working with the errors which are kind of almost uh and i'm not making a little bit of a jump here but u_m are kind of implicit to the to whatever the system of uh interpretation is on the data it is quite u_m it's kind of

uh that's why you're going into this biological world because it's so bottom up in a way um because it can start just with um whatever the errors are in the nature of the um the system that's been set up for doing the sensing even forgetting action just the nature of how the sensing set up will start to generate the errors at some implicit level would that be a fair assumption

so anytime there's going to be a prediction of observations so yeah we don't need action here yet there's two options either you have a hundred percent predictive power

and your error is zero or there was some error so you're always in one of those two cases and which one are we usually in not having the total perfect 100 accuracy down to the you know 100th decimal point if for no reason other than that's beyond the point of diminishing returns or we're also trying to balance accuracy with model complexity so because we have these metrics like the aic the bic all these metrics that find us a model that's a balance between accuracy and complexity so it's not overfitting the cost of not overfitting of cost of having a simpler model is literally increased error because every new parameter that you add into a model you always explain more variance the first principle component takes the most variance of the data the second principle component takes the next most variance of the data every principle component on a data set is going to continue to eat more and more variance out of the data leaving less and less in the error term but it's a situational thing whether you need to explain 80 of the data or 90 of the data or 95 of the data but there's always going to be some difference between the prediction and the observation again unless one is in this super edge case where

it's perfectly known so yeah go for it

yeah that's that's helpful and it's also implicit there that if even if you in whatever you set that complexity and accuracy to at this very deflated level is is really helpful so the nature of the model will change the nature of the errors and basically the best it gets to

by the most it can reduce will vary a little bit as you change that model um and not making too much of an extrapolation but that that is a nice way to be able to unify a lot more complex and rarefied ideas and you know what is really happening um you can sort of drop back to that complexity accuracy um principle but here's some more sections from 4.1 towards the end of 4.1 on page 30 in the paper so yeah hopefully we've been representing this at least mostly accurately and then for those who want to go deeper that's why there's a paper and that's why there's like a continued discussion and development on these ideas because they they drop many little crumbs of well we've held this to be zero for simplicity and but they're right we can think of this as a predictive coding network in which all the error is initially focused at the output of the network where the loss is and then through dynamical minimization of prediction errors at multiple levels layers this error is slowly spread out through the network in parallel until the optimum distribution the error at each vertex is precisely the credit that should be assigned to causing the error in the first place that's pretty interesting it's kind of like let the chips fall where they may with respect to error and variance attribution dean

this is this is awesome so are they also are they also is the is the way the architecture is and the way the graph

the graph is set up as kind of a mirror of what the architecture is of this layering thing
are they also there to kind of contain little fires of mistakes and errors is that why it's effective
because it sort of contains something you know suddenly causing a chain reaction and having and having
everything and when we read magid's paper about i think it was maybe it was axel's paper about you can
you could make predictions going right off of a cliff and and the result is you no longer exist is this kind
of an architecture that helps us avoid some of those really consequential prediction consequences
that's an interesting idea i think we should engage in some slide play and try to
try to get a little bit of clarity on that and like also hopefully we'll hit on um
a lot of the ideas that we've been bringing up in a more abstract way because i know it it's helpful
for like everyone to see multiple coats of paint on this and like it's the the epitome of abstraction
or that is the austerity of those formalisms about compute graphs so let's think about a um we won't take
it to the um the cafe level um that was even though that was i remember that being quite fun um with
the we're gonna be we're gonna be between the cafe um and uh you know something else so we'll be
intermediate because we're gonna be talking about a specific graph but we're not gonna like necessarily
for the first pass on it go all the way in okay so the node that we care about being most accurate on
is seven so like this is gonna be like our our focal node this is like the one that we want we want to
have an image classifier algorithm this is the classifier node we want to have um prediction on what
the thermometer is going to say that's what this is we want to have prediction on the temperature in the
room that's what this is we want to have inference on action and think about the upstream parents
of action then that's what this node is so seven is going to be like the node that we're interesting
uh interested in minimizing our loss function on okay
this can be carried out on multiple variables but we're going to just look at minimizing on this
okay what is the back propagation chain here well let's think about what the forward and the reverse
arrows are so first just to clarify like this is a directed acyclic graph there's no loops in the graph
and the arrows are all directed um they reflect differentiable functions causal influences amongst these
uh variables these intermediate computational products so like variable one is then there's some function
differentiable function that translates it into two two influences four and three three influences five
and six six influences seven okay we've seen this in the bayesian context where the sparsity of the
connections allow for the factorization of this graph like it allows us to sort of
hold one part um unchanging and change other parts through factorization that's a lot of the variational
inference stuff that we've talked about okay so what are the parents of seven which nodes do you think
that we should consider if we're going to back propagate so the forward model the actual like
causal chain in the model not in the world is like one you can imagine some perturbation happens to one
every node is going to be influenced that's the forward perturbation in the forward generative model
what is going to be invoked if we start at seven and do a back propagation
which nodes are not important five and four right yeah five and four will not be invoked in this optimization
scheme so if we were so we'll make these ones gray that we're going to be back propagating on
one but um and i i also hope i'm representing this accurately but um you could imagine that like

changing the dials on five or the function between three and five will not reduce the loss on seven so it still might be an important thing in the forward model but it's not in the back propagation from seven it's in the forward propagation from one um okay so then um this is not a predictive coding architecture because in the predictive coding architecture some of these um that's what we saw in like for example in figure two is what we saw in figure one so there's the means and the variance estimators so let's bring in figure one and try to adapt it to our graph so our graph is sort of up to formalism 29. okay now we want to think about a graph that's not just nodes and edges of the abstract type but it's going to be nodes and edges that include these error neurons as they're calling them here okay so here's a reminder on what that architecture looked like

jacks wrote how is the back propagation of error related to um uh holland's john holland bucket brigade the bucket brigade in signals and boundaries it sounds explain a little bit what is the bucket brigade or will look but it sounds like a bunch of people helping and sharing buckets or something like that um okay so let's just say that the one that we actually want to minimize is going to be like a μ it's going to be a mean estimator now let's just modify this graph so that this one's going to be error i'll i'll just use the um well type the variables so this is a mean estimator and then we're going to alternate so that this one is an error and this one is a mean this one's a mean again

gotcha this is not the exact same architecture that they laid out and this might not be um traditional or um classical standards but like this is getting us towards this idea that within a cortical column so to speak we have um the forward propagation of alternating layers of means and errors we want this loss function to be minimized by diffusing the uncertainty across these error nodes that is like letting the chips lie in terms of the variance attribution and as they write precisely the credit it should be assigned to cause the error in the first place so if like node 2 in the forward causal model was causing 90 of the variance and so this was like like one was you know is a very tight distribution and it gets blurred a ton on the way to three and then from three it only gets blurred a little bit on the way to seven so like two is going to be big this is like a big error estimator and six is small

tiny variance estimator so it'd be like if we were in classical statistics and we were going to do um you know we were doing brain imaging on a bunch of people from two different categories like with and without this diagnosis you could imagine that in that two level model one world there would be like massive variance amongst people and very little differences between those two higher groups in another world it could be the opposite there could be big difference between the two groups but very small differences but amongst the people within a group and so this is the parameterization and finding the parameters that are going to minimize the loss function means that we want the variance to be partitioned appropriately across error terms just like we would like the average the mean estimators to be accurate

the equivalent in the error world is we want the errors to be appropriate in how they're distributed so it's like it's like if you're doing a mean estimator on something you don't just want like the highest mean possible you want like the most accurate mean possible and the most accurate error possible doesn't mean the lowest error possible that would be the fallacy of well we're estimating means well then let's just make it bigger so we're estimating errors not because we want bigger errors or want smaller errors that's the thumb on the scale estimating errors accurately entails appropriately distributing the error terms on this graphical architecture and then they're adding in one extra point which is basically that the dynamics of predictive coding are purely local requiring prediction errors from the cortex the current vertex and the child that is on the computational front a win for tractability and for the actual algorithm because you don't need to have the whole model loaded into ram for example because you might only you know which neighborhood of variables you're going to engage when you're doing a certain parameter update and then on the biological side it starts also leaning towards plausibility because like they even call it an error neuron well neurons are signaling forward let's just call that the direction that the action potential travels down the axon but there's also retrograde signaling at the synaptic junction so it's plausible at the multiple neuron architecture level or even at the two neuron biological system level that there is like even if there's a direction at which signaling is assumed or perceived to be normatively happening there's also a retrograde signaling

of multiple kinds of multiple kinds we won't go into that like complexity of retrograde synaptic regulation but that at the circuit level and at even the synapse level the idea that signaling is happening both ways is something that is seen as a strength of the predictive coding architecture again the locality allows it to be more computationally implementable and this sort of bi-directionality and signaling is more reminiscent of biological signaling processes synapses conversations brain regions so on

okay let's okay yeah any thoughts on this

i think that that's pretty good explanation go for it steven just just checking so if you added all the errors are like are the errors kind of independent of each other or would they is there like a total error across the the chain so to speak of like uh how much error there can be or is it like kind of contained within the street like is six relatively contained and two relatively contained

the evidence

okay dean can i take a guess steven yeah sure i'm get i'm guessing so just keep the wild a money money part of

this front of mind um i think what the i think what the graph is telling us or at least when we we formalize it in this way is that you could always find the cumulative error but the point of the system is to try and keep the error where it actually exists within each layer so sort of sort of partnering the error the error at each level with with whatever the prior or the hierarchical I plus one or I minus one layer is but me again i'm just guessing but i'm kind of making sure that i understand completely what what daniel laid out here and that would be my guess i mean you could you can add it all up but i think the i think the point of it is is that before it it runs away from you this architecture is

set up so that there will be some cumulative effect but it probably wouldn't be it shouldn't be there's the word i'm looking for it shouldn't be as abrupt as if you didn't have the layers and you didn't have the ability to differentiate and you didn't have a kind of an understanding of what what is happening i.e back propagationally now again i'm just throwing that out there but because i don't have a reputation around this i could be completely wrong there's a lot to say and the authors like do provide many citations but i i believe that's right like just as appropriate mean determination we're thinking in a gaussian world like um there's a lot to say on gaussian processes but a gaussian is a mean and an uncertainty so it's there's going to be somewhere where the hump of that distribution is and then there's going to be some width to it now sometimes the process is actually gaussian other times we can use nested gaussian processes so that it's like well there's some process that's moving it and then there's wiggle around it so we're going to see that moving as one gaussian and then the other one is like a ripple that's like a smaller gaussian on that

and then even in cases where there's a non-gaussian distribution that's where the laplace approximation comes into play which is the second derivative we have the jacobian and then the hessian matrix in 26 and then we talked about how when you take the second derivative you're basically making that bowl and then lying in it so you're going to the bottom of the bowl that you created and you know that it's going to be the sort of hump structure that's amenable to going downhill because once you go beyond quadratic beyond the hump anything that's like a multi-hump distribution or even just x^3 you're going to have challenges getting globally optimal from local optimization processes because you won't know if you're in a local energy well or not and so many of these computational tools and approaches that we're discussing like across these live streams and papers can be understood as taking something that isn't like a bowl and making it so that it can go downhill and finding well what is the appropriate level of how many layers have to be added to this onion or how many oven gloves do i have to put

on how many approximations should we carry out in our approximation science so that there can be something

like a parabola where we know that we're on parabola territory and that's what allows us to then minimize a loss function because if one believed that they were on a rugged landscape then minimizing something locally would be you'd have no way to know if that was on the right track or not um yeah steven and could i say then would i be okay to say that it's almost like each of these areas say number six number two in here and you could have those would be like little mini bowls or mini um at different and the further up the chain the bigger the potential um impact of the positioning in that bowl or whatever the the landscape is out towards whatever's been you know the mean that's been understood

okay let's look at figure one so this is a cortically inspired so inspired by the cortex and blue wrote dean i understand the architecture the same way error correction is retained locally so let's look at this semi cortically inspired predictive coding hierarchical system just like in the example with the two clinical populations doing brain imaging there's one world where the patients within the group vary a lot and the groups

don't differ there's another world where the groups differ a ton and then the people within each group do not have any variance amongst them so depending on how it's parameterized one could have a model where the um highest level has a major influence or not

and that's the whole question it's like in a linear model is it more important like the um β part the slope or the b how much it's shifted well if the regime of attention is on linear modeling as an architecture it doesn't make sense because they're both just parameters in a model now for any given data set it's an empirical question to what extent changes in the slope or the intercept are associated with changes in the loss function and that's linear regression here's an architecture that gives us an approach to finding loss functions but how do we know how many layers how do we know how many of these side connections to make

but there's a few ways to go about that one could just a priori model the structure of the model of the graph around something else like previous work or some inspiration from a biological system they might also be interested in this structure being a parameter of a model like and that's the whole structure learning and meta bayesian approaches and all of that but that's pulling back another level and so that's the the challenge of structure learning because we can get that loss function within a given static graph but that doesn't mean that we have the best possible graph architecture

it's like if you have a hundred pieces of um data biometric data on people and you're making a multiple linear regression to find out their risk for some condition do you just use a hundred that might be overfitting it might be too costly computationally so how many of those variables do you use there's a process for determining that what's at the trade-off between accuracy and complexity where it's fitting the outcome well the loss function on the condition that you're trying to actually predict but it's not past the point of diminishing return or using variables that are uninformative or for example if there's two variables that are perfectly correlated with each other um that's called multicollinearity and like you wouldn't want to use those variables

so that's the challenge in the linear regression world and this is like taking it into another space but it has more analogies than not which is why spm statistical parametric mapping that textbook is for those who want to engage vital prerequisite because it addresses a lot of these questions on model fitting including on dynamical systems and it puts it with six feet on the ground in classical parametric and non-parametric statistics dean this just raises the question in my mind i haven't answered it as to whether or not a mean is constructed or if a mean is actually sculpted or some combination of the two i mean i'm not being a min 2 guy i'm going to assume that it's probably both like you can have the figure but it's your understanding of of what the difference is between an edge and a and a and a directional link which you have to kind of carve out from the overall image but that's again that's one of these things now where if you get into this predictive coding piece i think the product looks like a construct but an actual but in actuality there's there's got to be a huge element of this of being able to isolate the mistakes so that they don't overwhelm and if you're isolating something if you're removing some factor you're actually sculpting as well so maybe that's a bit of a philosophical overlay on the sort of mathematical operational side but you know me i'm always trying to turn

it into what's the what the practical function of this yeah it's like statistics as world building because if you just want the mean then just take the grand average if you want the mean and the variance there's a method do you want a variance on the variance how confident are you that it's 10 plus or minus one is it 10 plus or minus one plus or minus 0.5 or is it 10 plus or minus one plus or minus 0.01 so a tight estimator on the error as a parameter and so we're like playing with which parameters are seen as like the things in and of themselves the means and which ones with respect to means have this interpretation as being error neurons or being variance descriptors now one properly fit model which you could take into the you know 50 stacked levels 10 plus or minus one and then on that one variance estimator we have plus or minus 0.1 and then how how sure are we about the plus or minus 0.1 0.01 okay now how sure are we about that 0.01 0.001 that would be like a well fit model where as we kept on adding uncertainty parameters we would be making appropriate estimators now what would be like a malfit model and i hope this is accurate 10 plus or minus 1 and you go how sure are you on that 1 plus or minus 50. it's like the variance just exploded it's like well but you're saying that the variance could be like 0 or the variance could be 50 how much confidence should we then have in the mean estimator and so that's what this appropriate distribution of uncertainty is now let's think about node 7 an uninformative node 7. we're predicting temperature but it's just it's unchanging we don't have a lot of information there to fit a world model because it could just be a single level estimate it's 10 and it's not changing but as we deal with progressively more and more nuanced and informative things that we want to minimize loss functions on like natural language images from natural scenes action including the unknown consequences of real-time unfolding action those because the uncertainty becomes higher and the information content of what you're trying to minimize your loss function on is higher it um um not authorizes or licenses but kind of moves us into potentially having higher levels of model complexity as simple as possible and not simpler with that principle being carried through again if you're just trying to predict a constant number you don't need a six level predictive coding architecture whereas if there was something that did actually have that many levels of depth that's the kind of architecture that would do well to predict it and we talked about this with the par and pazulo on the architectures for homeostasis and allostasis and it was like yeah to just recognize if you're out of bounds and then come back in that's a given model but then there's a different graph that's going to do something like intermodal or anticipatory or have memory so all of those cognitive or computational features license or engender more complex causal models but these are being sculpted ab initio from nothing by the statistician and then we're in sort of brackish or gray waters when we um juxtapose the architecture of the graph with the architecture of some natural system okay yes steven yeah and you've got this and you've got this excitery and inhibitory dynamic going on um and even though the arrow is going from the arrow down to the u that there is a the the u is exciting the layer above isn't it it's exciting it's almost like it's like one bottom up is kind of exciting higher up areas and then things that are further up also

sort of saying hey calm down calm down i can reduce some of this you're getting a bit out of hand here you know they've only scored two goals let's relax you know that that again i also hope this is not uh inaccurate it's almost like um there's a suppression of error like a dampening being carried forward with the ultimate dampening being you predict it perfectly you have totally quelled all error yet in real settings because of just simply model simplification or uncertainty in the world error is coming in from the observables which is often what we're trying to reduce our loss function on and that's this whole difference between reinforcement learning finding oneself in reinforcing or rewarding states

and active inference with reducing surprise about outcomes we're dampening error and carrying that forward in the forward generative model and then also able to run it back in a tractable way because it's like you're you're you're dampening the vibration but then the vibration is still entering in and you'd want that vibration to be appropriately distributed according to where it should be okay dina then steven yeah it seems to me that the way what this is what this is saying to me is that because of the way things are organized um two two things can live simultaneously at once one is sort of that local uh mean correction and the other is the more generalized global or the other direction in the bi direct bi-directionality parallel of of mean correction one is kind of reinforced through external evidence while the other one one is reinforced through localized almost almost it is distributed but it's almost like the ability to localize and isolate if necessary so i know we've talked a lot about top down and bottom up but this brings a lot more color in my mind to some of those conversations we've had before in terms of you know the arrows going in opposite directions and it's not so much that they're going in opposite directions but that you can be going two different directions quite literally what literally at once because of the way things are set up so yeah i think this this has been really really helpful for me in terms of not only understanding what back propaganda propagation is but actually in sort of giving me a little bit more foundational ideas around again how do we turn this to these sort of statistical ways of representing it into so i'm confused should i should i go into the cafe or not i mean we do not not all of us have to go to that level of detail to really understand what the final thing was that pushed us over the door threshold or not but it appears that it's this has been around for a while and being able to take that if you really are kind of stuck because you really don't know what the the parents are of your current action because we haven't got as you say we haven't got to action yet but that's the whole point there is an architecture there that's going to allow us to have choice

thanks um this minimum of two um comes up so much so it's awesome to hear about that and then to connect

like expectation maximization and the tale of two densities epistemic and pragmatic it's like wherever we look there's partitions into two into multiplicity and unity through plurality and then other times even within the one road there's the movement in both directions forward and backwards in time or the forward propagation of a generative model which could from just one if all you knew was one you could simulate sevens if all you knew was seven and the structure of the network of course you could back propagate the best possible mean and error estimates using back propagation of error okay

stephen this is helpful for thinking about how we perceive things as well for i was dean was saying about making choices it can be when we get zero error or whatever seen as negligible error you want to just allow that to carry on and stay with your imagination so i go to the bar and i a lot of things are the same right maybe there's a happy hour sign and if that's unusual maybe i'm happy about that i note it um but if a lot of things are exactly or very similar to how i expected then in a way my perception is probably mostly my predictions just not being contradicted it's all oh they actually i just don't see it i just i think i've seen it but i just don't need to see it and where there's something which is like ambiguous like ambiguous like you were saying you need that number seven if we're going to use this diagram starts to have to be looked at you know you start peering at things straining their eyes trying to sort of see well what's that is it is it a dog is it a man running into the woods is it a dog is it man running into the woods um you you you know you start to notice things so i think this is also quite useful for that perspective of you know when when you just you don't actually need to see a lot of stuff it's quite an efficient model right if it's if the if the area goes below a certain level you just move on and carry on of course that's where the challenge with autism is they think because too many things have been looked at that are possibly negligibly relevant at the time okay thank you so just one last one last thing down before because i know you want to i clean this up i think then one of the analogies that i draw is with the emphasis on the word minimum to min to doesn't mean it's always a you know a hard bifurcation and or a binary it means that the minimum amount of things like even when you take your shoelace and wind it through all the eyes of your shoe you've got two ends that you eventually will loop back together and and entangle in some way to hold your shoe on so even though it's one lace i think what a lot i think what active inference reinforces because it's active and inference is that ability to be able to see two things both seemingly contradictory and both actually true now it can be multiple things you can have multiple as as you can have multiple surprises some of which are make you quite happy and others where you walked in and said well why hasn't it been happy hour at this hour when i've been at this bar on multiple other days like you could be disappointed in that as well as being pleasantly surprised so i think what this again i just reinforce this the the minimum of two allows for the integration the minimum of two recognizes the differentiation and when we're talking about statistical issues and when we're talking about errors and when we're talking about energy um i think i think the way that they that these folks have been able to discover what so what architecture would allow for that is what's really interesting to me now because i i wasn't even thinking in these sort of so what's what is the what does the scaffold look like but it looks like there's been a lot of work done in this area so so let's take it to action as we head into our last little bit on this and of course if anyone is more author or not familiar with the formalisms just contribute to act and flab help us annotate the papers and read these deeply because there's so much to learn and somebody getting involved like a few days or weeks before these streams is a leverage point if you want to help other people understand this area so i've changed the colors just all to gray

because they're all just random variables now we had the seven so we were back propagating error on temperature prediction but also maybe this system is predicting um pressure and so then three might be like estimating you know is the water boiling again just we're getting one one step closer to the cafe so if if we were only trying to predict temperature we would back propagate along the seven six three two one route if we were only predicting pressure we could back propagate along eight five three two one the fact that there's a parent that is upstream it's like the last common ancestor of pressure and temperature like we might be able to reduce um these errors a lot on the temperatures by having this higher order variable is the water boiling and so when can mention like if the answer is yes then it's like it's expecting that this is like

higher on both and if it's low it's lower on both so the generative model can go from the water's boiling what pressure and temperatures do i expect and you can add a third node it allows you to condition on a cognitive structure go from observations of just temperature around the horn to pressure or from pressure and temperature up to the water boiling but the influence of these observations is like limited through this choke point it's actually a markov blanket with respect to what's on the either side of it

let's bring action into the picture action can also be understood as parametric

what angle should my elbow be at what angle should the eyes be oriented

those types of things again not whether that's what the actual neural signaling in the brain is doing

but this is how we model action it's like the the car how many miles per hour should i be going

those kinds of questions is parametric modeling of action and that's what was so fun and exciting

was the first 50 equations in this paper have scaffolded this tremendous general node computation

architecture. Multi-level, means and variances, forward and backward, up, down, left, right,

all of these interesting ways and connections, but not yet action. Not because we hadn't brought

action in as a dancing partner, but actually this was above any interpretation of the nodes as

inference or action. And then it's as easy to condition on action as it is to condition on any

other inference, because action is just a parameter for parametric active entities that are engaging in

action as inference, planning as inference, and so on. And so then one can think about what is the

compute graph, whether it's composed of just nodes and edges, or whether we think of some of those

nodes as being like variance estimators. And then, you know, this, maybe, again, the connections,

I'm not going to rewire this whole graph, but just like, is, is this a water boiling situation?

Okay. So we have a thermometer. It says that it's a hundred C it's boiling temp. We didn't,

the pressure sensor is broken, but we back propagate it. We're in a water boat. We're high confidence

on the water boiling. Then this is like another variable that's going to require like a different

sort of question that it's answering. But one can imagine that if this is a water boiling situation

and the water is not boiling, there might be an error. And so how can we reduce loss function

on the total compute graph, including observation and action? How can we reduce uncertainty about a

cognitive model, including observation and action? There's two ways to do it. And it's how they

introduce it in the very beginning of the, of the paper, which is that the minimization

of the loss function of the free energy, expected free energy, variational free energy can be achieved in multiple ways through immediate inference about the hidden states, the world, explaining perception, updating a global world model to make better predictions, slower parametric inference updates that have the interpretation of learning. And when action is a variable that we have agency over, it introduces some complexities like the need to have preference and the need to specify a time horizon and to have uncertainty about the consequences of action, right? Wouldn't it be easy if we were just watching the movie and our actions didn't have any influence upon the story that we saw? It'd be a different inference. And finally, through action, to sample sensory data from the world that conforms to the predictions, potentially providing this integrated account of adaptive behavior and control. So it took us 50 equations at the general before we could jump into action.

But it's a deflationary approach to action. Because action is a parameter in a broader cognitive and computational framework. Okay, Stephen?

Thanks, Daniel. Yeah, that's really helpful. Actually, going back to that, the example with water as well, you can even bring in the higher level of affect, in the sense that one thing that's really hard with water, compared to a lot of other things when you heat up oil and stuff is, you can have the water boil, and you've probably had it. And you can have a thermometer, theoretically, say 100, but the steam can go above 100. But it's very hard to really, it can trick you, right?

It's like, it's hard to, it's actually quite hard to get, I find it hard to get my head around steam being hotter than water, because it's kind of, just kind of how I know, you just think of it as being 100 degrees water, right?

And it's almost like, if you were to get into that situation, or after I burnt myself one time, or there's almost, maybe I just start to have a feeling of fear, if I have a sense that I'm somewhere where, what might be coming out is high temperature steam, right?

And that will override the fact that, perceptually, I haven't got a lot of ways to easily, sort of, gauge that visually, unlike, say, a ring getting red, and, you know, other things.

Okay.

Okay.

Here's what that makes me think of.

Sometimes, we have indirect cues.

That reduce our uncertainty about something that's not directly observed.

Like, the mood ring.

If everybody was wearing a mood ring, and that was an accurate predictor, you know, that would be this other alternate world.

And so, there are certain kinds of variables that are more like the magical mood ring, or like the temperature of metal within a range.

Because, I mean, it's only going to start to get glowing when it's super, super hot.

So, it doesn't help you differentiate whether it's free, like at 0C, or 50C, or 200C, probably even.

But there's some range for certain things in the world where we can use indirect sensory cues.

Affect is not quite here in this model, in this graph.

But we've talked a lot about how uncertainty can be understood as like anxiety and or negative valence.

In the world of reward maximization, less reward is bad.

In the reward, in the world of precision optimization, excess variance is bad.

Not any variance.

It's not a simple, you know, destroy the variance estimators.

It's about the appropriate applications of variance.

But excess variance, more variability than expected.

You told me it was 10 plus or minus 1.

But I just pulled 3, 50, negative 80.

That's confusing.

It's more confusing than I thought.

So, then some of these mean and error estimators in a computational model might have...

This could be a mean estimator on valence, or one could choose to write a paper where an error estimator is framed as being an anxiety parameter.

And that's always that jump between what any parameter is in a graphical model and anything about the world or attaching it to some word.

Because, oh, it's anxiety.

Well, then that makes me think of this other situation.

So, you can send someone really too easily down a road of associations when the deflationary perspective is just stating what it is.

Okay.

Okay.

Fun.

Didn't expect that we would go this way with the back prop, but Blue mentioned it last time, and it is important to bring up.

And also, it connects to modern methods for training neural network architectures.

And that's what a lot of the author's research has focused on.

So, let's just continue to explore action a little bit.

They reiterate that when we're thinking about active inference, about inference involving action.

So, cybernetics, control theory, any area where some parameters represent observations or latent states, and other parameters represent active things.

The first way that you can minimize prediction error is to update predictions to match sensory data corresponding to perception.

And the second is to take action.

And earlier, from the introduction of the paper, they kind of did a little minimum of two even on this, which is to split out perception as rapid inference and learning as a slower updating.

But we've talked about the continuity between perception and learning.

Like, if you're seeing the ball move across the screen, is that perceiving the movement of the ball?

Or is it learning a parameter representing the location of the ball?

So, whether we call something more perceptual or more learning-oriented,

and even abstract learning can be seen as a perception in interior spaces, which has been the focus of a lot of work by Fields and Levin with competency in navigating these abstract spaces.

But perception sometimes brings us a little bit, not too close, but a little bit just linguistically to, like, our experience of perception, when the phenomenology and the qualia is not what is in play here.

It's about parameter updating in the model.

Yeah, Dean?

Do you think, Daniel, it's about sample size, but sample size is related to maybe the time commitment?

Like, if I'm inferring something, that's a very, very short volume of sample versus prediction, which is slightly or relatively larger volume of time commitment to move to that volume, and then modeling, for example, which takes the perception and tries to build something material from that? Is that, do you think that that counts here?

Because at least that's what I was, that's what I was telling participants we had to take under consideration, was sort of the time factor of the sample sampling process.

Sure.

I think if we allow inference to be just broad and including everything, then that's what it is.

But just in its connotation, inference using a model often has the, it's sort of like the machine is ready, the model is ready, and we're going to do inference.

The new patient comes in with a biometric data, we're going to do inference using this model, and we're going to predict their risk of the diagnosis.

Okay.

That's the fast perceptual timeframe.

Learning is, unless it's one-shot learning, is taking in multiple patients and updating the parameters of those models.

Now, those two stages can be alternated like in expectation maximization or in the tale of two densities, but that doesn't get away from the fact that there's sort of this like one data point mode, like plug and chug inference perception mode, variational free energy.

And then there's a little bit more of like the updating the model parameters.

It's like, are you tuning the engine or are you using the engine loosely in the non-action space?

In this first way to minimize prediction error.

Given the ongoing stream of patients coming to your office, there's two ways to reduce your surprise.

Those are the perceptual and learning continuum, which does come down to the specifics about like the sample size,

like you're in the architecture of learning and perception.

You could also reduce surprise by choosing what to sample and or modifying the world.

So that's how they bring action in.

And it's very interesting to think how many pages of prose have gone into wondering where action fits in.

And there's still many more pages to go in terms of where does Formalism 51 take us?

Have we cached the check of pragmatism, inactivism, extended cognition, and so on?

Like what is 51 with respect to all the qualitative insights of action with all of its depth?

What is 51?

A short answer might be that it's like the y equals mx plus b with respect to all the ways that linear models are used in the world.

This is just like a sort of skeleton archetype that is just the trace or a hint

for someone who then wants to analyze specific actions over specific timeframes for specific cognitive computational models.

But that's where 51 takes us.

Stephen?

Yeah.

One other piece that could be kind of interesting to throw in there,

as well as matching my sensory data with my predictions,

I could reimagine my reality to explain my beliefs,

which in a way would mean I create my own sensory data by proxy, if that makes sense,

which is a kind of an interesting slight add-on to that.

But it's the same idea because you're not strictly taking actions.

You're taking actions to match sensory data, but not necessarily the real world sensory data.

It could be the kind of, you know, the imagined world data.

I am seeing an angel coming over the top of that hill now, you know?

Yeah, one could engage in mental reverie,

and it would either be reducing their surprise in the long run,

or that system would fail to persist in an entropic world.

And people have made all kinds of arguments, like in evolutionary psychology,

about the basis of why we perceive or believe different things along similar arguments.

So, definitely makes sense.

Let's just look a little bit in 452, which we didn't previously get to.

But, of course, anyone with any other questions in the live chat, welcome to do so.

So, here, in 26, livestream 26, we talked about PID control,

we talked about integrator chains, and about generalized coordinates of motion.

So, just a reminder on what 26 looked like.

We had, from left to right, is time.

These are different time points.

And then, at each time point, there's this vector, it's like a stack,

reflecting the observables.

And then, also, things that are latent states,

they're not observables of the world, real or imagined.

They're the derivatives of how the observables are changing.
And those can be the position, it's like the zeroth derivative,
but then also velocity, jerk, so on.
Acceleration, so on.
So, we looked at that in 26,
but it wasn't connected formally to predictive processing, predictive coding.
PID control is restated in 53,
using the formalisms, using the notation that is going to prepare for the merge with predictive coding.
Like, A for action, epsilon for errors.
And so, the error term is included in all of the three levels.
To obtain equivalence to predictive coding,
we utilize a linear identity generative model with three levels of dynamical orders.
So, the joint distribution of all the O's, all the observables, and all the X's through time
is going to be this factorized expression.
So, the distribution of the observation conditioned on the latent state,
and then that is conditionally independent of,
so it allows it to be factorized and calculated out separately.
Here's like the first derivatives, second derivatives,
with O double prime and X double prime.
Here is X conditioned upon its own derivative,
the derivative of X conditioned on its derivative,
and the sort of root node is the double derivative of X.
Because this one like stands alone, that's like where our model kind of taps out.
That's equivalent to node one here.
Like, there's no variance term leading to one.
There could be, but then that would be a different model.
So, these equations in 54 restate this dynamical model
and introduce omega, which I believe would be an error term, this W,
and also the mu's, which is the desired set point for X at that order.
So, at the first, second, or third derivative,
what is desired there,
then that allows,
so this is like a graph where instead of these being different things,
like what if, what if,
kind of mixing architectures a little bit,
because the generalized coordinates doesn't have errors interleaving.
It just has the state, the first derivative, the second derivative, and so on,
rather than interleaving error terms.

But we can imagine another similar graph,
where as you backpropagate,
it's like higher and higher derivatives.

That's the integrator chain.

But, I guess, in 55, 56, and 57, and 58,
they do something similar.

To what we had done with the generic computation graph,
or to the predictive coding architecture,
with the interleaving of the means and the errors,
this is just that,
on the graph architecture,
where the nodes have this interpretation of being derivatives
derivatives of one another.

What is there to say?

That's an interesting introduction.

Just bringing back flashbacks to trying to,
trying to get my head around 500 level economics courses,
and when you'd have somebody walk in and start pounding stuff up on a whiteboard,
on, you know, second, third, and fourth derivatives,
and just feeling like you'd been tossed into a blender,
and you had no way of avoiding the blades.

So, I'm glad that there's people that can do this stuff,
but I would have to dedicate way more time to trying to parse what the heck they're talking about,
because it's a wall to me.

It's really hard.

Yeah, I feel you.

Maybe for the last few minutes, we can kind of pull back to,
okay, as we swoop out of this bathtub,
out of the dot two,
into the great beyond,
like,

what have we taken with us?

Why was this lengthy journey through predictive coding,
and the 60 formalisms,
and what Maria brought with the thousands of years of philosophy,
and history on perception?

So, where are we,
and how are we different now,

and going forward,
than we were before we started 43?
I'll say what it's done for me.
First of all,
bringing up the historical piece of it was really,
really helpful,
because this didn't just sort of come up yesterday,
or in the last couple of years.
It's been around for a while.
I think,
for me,
predictive,
I wouldn't have seen predictive coding
as a sort of formalized way
of retracing a navigational route,
but that's what it screamed to me,
and after the point one,
I listened to you,
hearing you and Blue talk about it,
when the idea,
when the question around backpropagation came up,
that's when I sort of did a deeper dive on that,
and again,
it wasn't,
I wasn't picking up on how to do that path retracement.
The formalisms weren't really addressing
what I thought was being said.
I think what today helped me with is
sort of examining,
if you don't have traces laid down
that are material that you can see,
if you weren't actively part of the building of the path,
that there's still some graphical,
statistical ways
of organizing information
to give you a better sense of,
okay,
so you didn't lay the breadcrumbs down,

but we can still give you a better sense
of where you came from
as a result of being able to use
this kind of formalism.

I mean,

honestly,

Daniel,

53,

53?

Come on.

I mean,

if you're one of the people
that was able to write those 53 out
and actually present it
in paper form,

I mean,

props to you.

You get,

you get many thank yous,

but

holy mackerel.

Other than the fact that,

that there is some evidence now

that you can retrace,

even though you don't have necessarily

those markers,

or

you didn't self-correct,

you didn't,

you weren't aware of how you self-corrected,

whereas

the predictive coding

at least gives you an opportunity

to maybe trace that,

retrace that back,

even though you didn't spend

a lot of time trying to

organize the evidence.

The other thing that it does
is I think it reinforces the idea
that the way
that these live streams are structured,
which is just introduce the paper,
just give the paper its,
its opening,
like open up the box
in the point two,
you can actually go
into one particular area
of the unknown,
which for me
was the back propagation,
which you and Stephen
helped me with.

In the point two,
we are still doing the part
where
the point two
could always,
even if you didn't have a record
of the point one
and the dot zero,
this paper told me
that you would still have
some way of knowing
where that
elusive dot zero
exists relative
to this live stream today.

And again,
I could be overfitting,
but I don't think I am.
Otherwise,
why would you
spring out 53 operations?
A lot of interesting stuff there.

So,
so,
um,
because this is level,
this is the exact,
I mean,
what we're doing in the live stream
and the levels
that they're talking about,
they,
they affect each other
in exactly the same way.
It's diagrammed out here
as a stack.

And we tend to think of ours
because it's over multiple,
multiple days
as sort of a horizontal,
horizontal continuation,
but they're essentially
one and the same.

Oh,
that's,
that's very interesting.

Um,
so it's almost like,
well,
there's,
the dot zero
and the one of the two,
they can go both ways.

So in,
in the temporal sense,
the dot zero
is a precursor
temporally
to the dot one,
the dot two.

So for the,
for the,
um,
for the,
uh,
YouTube channel subscriber,
this is their observation.

Oh,
43.0.
Oh,
43.1 has been live streamed.

43.2
has been live streamed.

But,
um,
what is like
the derivative
that sort of sets
the initial conditions
for dot two?

It's the dot one.

Yeah.

And then what,
of course,
sets the,
for the dot one,
dot zero.

And so there's almost
this sense in which
as we,
um,

that high school teacher,
that book that you read
all the way back,
even beyond one's own
corporal birth
as like

parameters that
dampen
or specify,
if one doesn't want to think
about dampening,
the position,
acceleration,
and velocity,
and,
and so on.
Um,
they're like higher
and higher chains.
And that means
that they have like,
they're further and further away
from where the rubber
hits the road
for the YouTube subscriber
on the outside.
But they're absolutely
like part of the process
by which something
comes to be.
And then the dot two,
you said,
can exist without the dot one.
And then,
and then the first time
it made me think of like
the negative numbers,
like even before
the zero,
what is,
what is there before zero?
What happens before zero?
Is there a negative number?
There's no number

that's positive
that's smaller,
but what is,
is it,
is it a letter?
Is it a shape?
Like,
is it its own big bang?
It's like,
and there's a sense
that it is,
and there's a sense
that it isn't.
Um,
that's kind of.
Part of it's the pre-reading.
The other thing
that I think
maybe doesn't
sort of register
explicitly
is that
the error propagation
in the way
that we do
the live streams
remains.
There are some corrections
and there are some,
there is a sort of
a gradual movement
towards a bit
of minimization.
But every time
we,
we come together
and try to figure out,
we don't come

from a position
of,
we figured it out.
We come from
a position of,
well,
what will we,
what will we do
with this today?
So,
we retain
that error propagation.
And I think
that that's a,
I mean,
it's,
that's not convention.
Convention is,
why would we spend
time on something
where we leave
a bunch of error
baked in?
But I think,
when,
when,
again,
if we're,
if we're going
to differentiate,
if we're going
to truly integrate,
if we're going
to be able
to say that
you can have
a point two
existing

without a point
one or a point zero,
of course you can.
But how superficial
is that relative
to the method
because of the
architecture,
because of the way
it's structured,
how much deeper
you can actually go
and how many times
you can bump
into a dead end,
immediately find out
how quickly
you're going,
or,
you know,
going off the rails,
so to speak,
because I can't remember
which one of the live
streams was that
you had the train
moving down the track.
But you can find out
whether or not
something is a dead end
quickly and recorrect.
That's what that
retracing part today,
I think,
is one of the,
well,
it's the takeaway
from this set

of live streams
that,
and I wouldn't have
anticipated,
based on the title
of the paper,
that that retracement
piece was going
to be the thing
that was going
to sort of percolate
and make the most
sense in terms
of why the structure
is,
yes,
there's an optimizing
piece to it,
but there's also
a piece that,
that allows for mistakes
without,
without,
you know,
I mean,
sometimes it's
very consequential,
but lots of times
it's,
the consequences
is what's minimized,
not the error.
And I think
that's what the
architecture affords.
You want to talk
about affordances.
So when you said

the consequences
minimized
and not the error,
that makes me think
about,
like,
high reliability
organizations.
Of course,
one way to minimize
the consequences
of error
would be to
never have errors.
Zero tolerance
for anything.
But that is
challenged
in a world,
again,
that's always
throwing us
challenges.
So,
in,
in,
in that setting,
it's almost like
the observation
here,
the one that we
actually want to
reduce our loss
function on
is not the
so-called proxy
of error occurring,
but we could reduce

our loss function
on function,
on the thing
working,
and just say
we are minimizing
the consequences
of all this
upstream stuff,
stuff that might
have the interpretation
of error,
and things that
might not.
They're just nodes
in this graph.
And then,
what is going
to dampen
surprise
about the
consequences?
And so,
like,
if you say,
you know,
anything that you
say to me right
now,
I'm going to
accept it
and believe it.
And then,
it's like,
in a,
in a trust space
that opens up
there to be,

like,
a communication
that goes deep
if it's being
honestly said.
And that's because,
like,
what's being minimized,
like,
don't tell me
something surprising.
it's like,
my consequence,
you can have
precision on
my consequence,
that's one direction
of the conversation,
and that is what
allows that
second direction
to really
come into play
with the
transmission
of
variability.
Rather than,
well,
how am I going
to reduce my
uncertainty about
action relative
to what the
person is going
to tell me?
I won't talk
to them.

Or I'll tell
them what to
tell me.

Or,
um,
I'll leave them
in the dark
about how I'm
going to act.
And then they'll
have uncertainty
about how I'll
act if they say
something even
trivial.

And so I think
this, like,
really,
it,
it spans many
levels and ways
here because
there's nested
levels and deep
chains.

but I,
I agree.
I think on
the more
topical,
um,
if such a
thing could
exist,
some of the
takeaways that
I had.
Also,

um,
really,
um,
appreciated
Maria providing
a lot of the
philosophical
context and
then that one
part where
we're like,
wasn't this a
meme like
thousands of
years ago and
in many
cultures in
the world?
What is it
that still
makes the
predictive mind
and surfing
uncertainty and
being you,
what makes
these novel
and surprising
and even
somehow
countercultural?
What,
what is happening
there?
What chain is
leading to that
where the idea
that we're

generators with
agency rather
than passive
receivers,
why is that
such a hill
to climb and
die on?
that's one
piece I
absolutely
take away.

Then it
was awesome
with blue,
um,
to work on
this in the
dot one
where we
had like on
the bottom
left,
the qualitative
philosophical
insights and,
and connecting
to technical
and formal
insights on
the bottom
right and
biological
systems.

And it's
sort of like,
where are we
engaging deeply

with one
of these
vertices?

Where are we
engaging deeply
with an edge?

Where are we
trying to grab
two vertices
and the edge
or go
around the
horn?

Or where are
we trying to
really go for
the triple play?

Um,
every baseball
home base
inference and
action,

um,
in the bottom
of the ninth
of the dot

two,
it's,
it's a
full count
and it's
a power
play.

That,
and then also
I think the
last piece
that,

the kind of
takeaway is
just when
the computational
and cognitive
framework are
defined generally
enough,
like at the
level of a
graphical
network,
which opens
up message
passing,
variational
inference,
back propagation
of error,
predictive
processing
architectures,
hybrids of
all of these
approaches,
then action
is deflated
or it's
inflated,
maybe,
or whatever
it is that
we even
say,
it becomes
composable
and tractable
to discuss

open-ended
combinations
of perception,
cognition,
and action
and impact,
which is the
home base
in the niche.
And so,
if even
that model
has to be
structurally
revised,
like we
end up
having another
way of
framing
entity
and the
niche
interaction,
it may
fit within
this framework
or it may
necessitate
revision of
the framework,
but within
the entity
niche
interaction
space
and the
nested

systems
and the
counterfactual
cognitive
systems,
there's
almost like
less and
less to
action
than it
seems.

It doesn't
remove the
challenge of
action.

It reduces
something else
that I'm not
quite sure
about what it
is.

Like,
it doesn't
make it
easy,
but
something is
being removed
as we're
learning and
working through
this,
moving forward
to think
differently.

It doesn't
make the

climb,
well,
it makes the
climb
different,
to think
differently,
but it
doesn't
move,
doesn't
mean that
fewer
foot pounds
of force
have to
be applied.
But there
is something
different about
the navigation
around action
and inference
via active
inference
that I
think will
continue to
develop for a
long time.
okay,
Dean and
also Stephen,
thanks a lot.
Wasn't sure
there for a
second at the
beginning.

I got to
thank you
again,
Daniel,
because
obviously you
did understand
have a
reasonable
understanding
far more
than mine
in terms of
what that
backpropagation
implication was
in terms of
marrying it to
the predictive
coding.

So I didn't
mean to put
you into the
tutor's seat
today,
but I
appreciate
the fact
that you
were able
to sort
of pull
it apart
and then
put it
back together
in a way
that made

a lot more
sense than
just reading
it off the
page.

So thank
you,
sir.

Thank you,
Dean.

See you
later.

All right.

Take care.

Bye.

Bye.

Thank you.